AI studio infrastructure design

Six people, 24 Claude and ChatGPT accounts, one hub for hill-climbing loops

24	$2,400	$1,200	$3,600
accounts: 6 people x (Claude + ChatGPT) x (team + individual)	individual accounts per month, paid by each person	team seats per month, billed annually, paid by the company	combined per month

What this setup is for

Six people share one hub that runs hill-climbing loops. A hill-climbing loop tries several changes to a piece of software, scores each one with the same tests, keeps the best one only if it beats the current best, and repeats. The hub holds the code, runs the scoring, and records every score. Claude and GPT accounts write the changes.

Every person has four accounts. The Claude Team seat and the ChatGPT Business seat belong to the company and do company work. The Claude Ultra and ChatGPT Pro accounts belong to the person. They add capacity when their owner chooses to start a round.

Infrastructure map

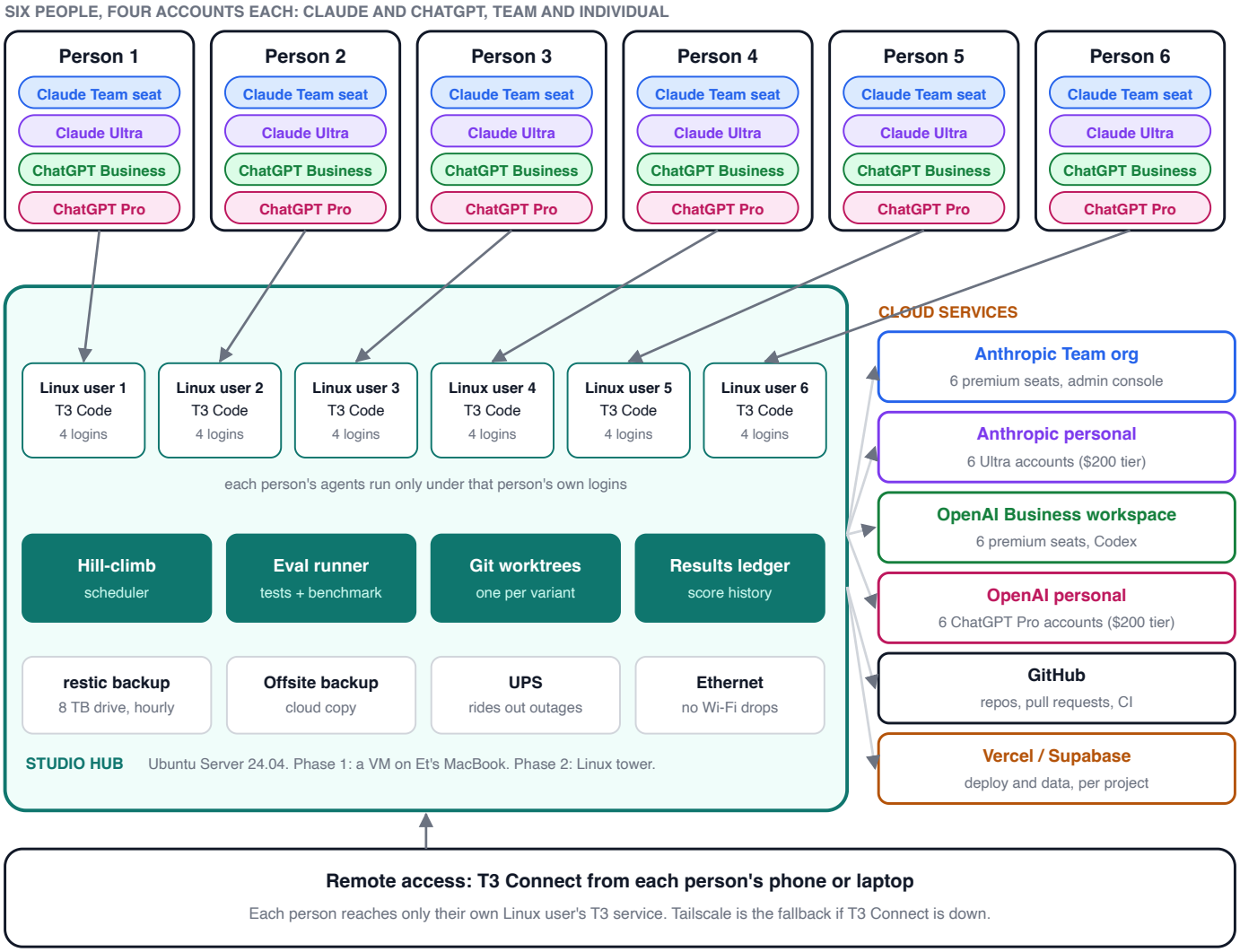


Figure 1. Accounts on top, the hub in the middle, cloud services on the right, remote access at the bottom.

Account rule. No account is shared, pooled or routed through a proxy. Each agent runs under the logged-in person's own account, inside that person's own Linux user. CLIProxyAPI and similar token-pooling proxies are left out because Anthropic's and OpenAI's terms restrict sharing and routing account credentials. As a company policy, jobs the scheduler starts on its own use company API keys and the Claude Team credit pool. Whether a subscription login may drive those jobs is still an open question for Anthropic, so the plan does not depend on it.

The hill-climbing loop

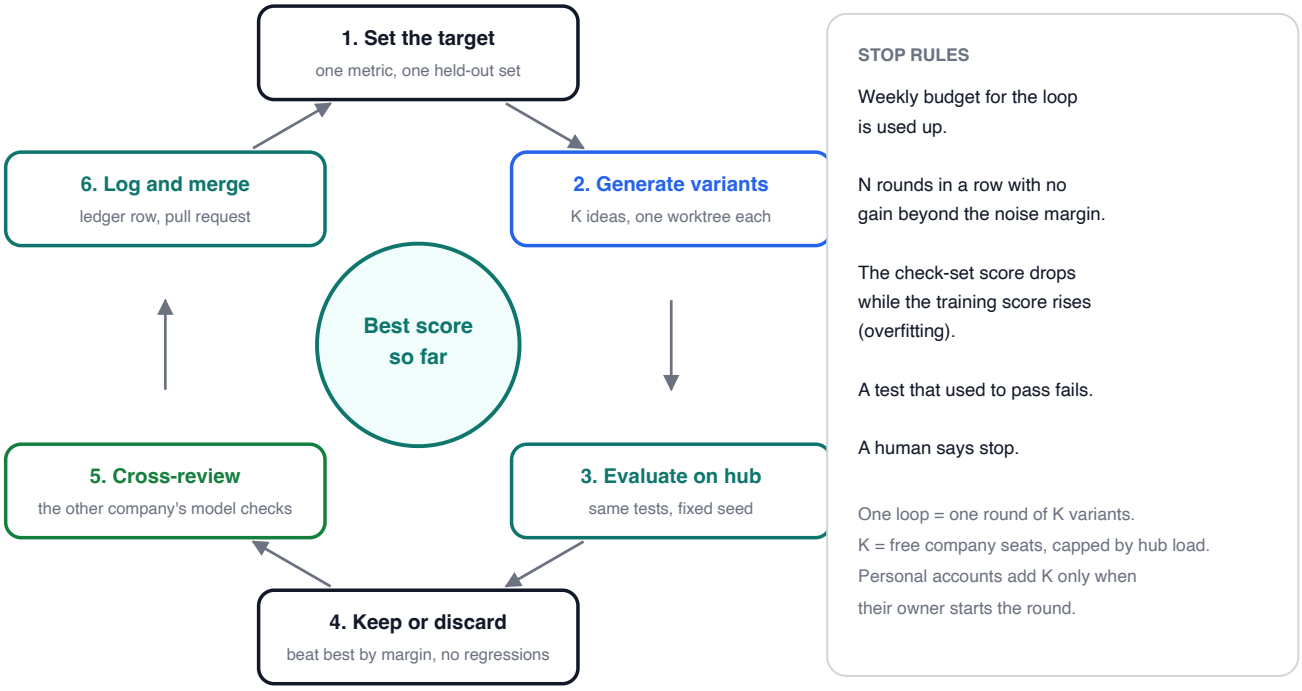


Figure 2. One round of the loop. The center holds the best score so far.

How a round works

1. Someone writes the target in a one-page brief. It names the metric, the smallest gain worth keeping, and three data sets. The tuning set guides the loop. The check set confirms each winner. The audit set is used once, at the end, and never guides the search. The noise margin is how far a score moves between runs of the same code. Measure it from enough baseline runs to see the spread, more than three.
2. The scheduler opens one git worktree per variant. Each free company seat, Claude or ChatGPT, takes one variant and writes one idea. Mixing the two models gives the round more varied ideas.
3. The eval runner scores each variant in a throwaway sandbox with no secrets and no network. The tests and benchmark files live outside the variant's worktree and are checked by hash before each run, so a variant cannot edit them. The runner checks that every expected test actually ran, not only that the exit code was zero.
4. The winner and the current best are re-run side by side, in random order, across several seeds. The winner is kept only if its gain beats the margin in that paired comparison and it breaks no existing test. For pass or fail tests, exact gates replace the margin.
5. A model from the other company reviews the winning diff: GPT reviews a Claude winner, and Claude reviews a GPT winner. It looks for a change that games the metric, for example a test that was deleted or a special case for the benchmark input.
6. Only the scheduler writes the ledger, one row per variant. Each row records the run, base and variant commits, evaluator hash, seed, score, failures and usage. Winners merge one at a time. Each one is rebased

onto the current main branch and scored again before the pull request opens. A person merges it after reading the change, not only the score.

Who does what

Account	Type	Used for	Who controls it
Person 1	Claude Team premium seat ChatGPT Business premium seat	Company repositories, hill-climb rounds, anything the company owns	Company admin (Et)
Person 1	Claude Ultra ChatGPT Pro	Their own projects. Extra hill-climb capacity only when they start the round themselves	Person 1
Person 2	Claude Team premium seat ChatGPT Business premium seat	Company repositories, hill-climb rounds, anything the company owns	Company admin (Et)
Person 2	Claude Ultra ChatGPT Pro	Their own projects. Extra hill-climb capacity only when they start the round themselves	Person 2
Person 3	Claude Team premium seat ChatGPT Business premium seat	Company repositories, hill-climb rounds, anything the company owns	Company admin (Et)
Person 3	Claude Ultra ChatGPT Pro	Their own projects. Extra hill-climb capacity only when they start the round themselves	Person 3
Person 4	Claude Team premium seat ChatGPT Business premium seat	Company repositories, hill-climb rounds, anything the company owns	Company admin (Et)
Person 4	Claude Ultra ChatGPT Pro	Their own projects. Extra hill-climb capacity only when they start the round themselves	Person 4
Person 5	Claude Team premium seat ChatGPT Business premium seat	Company repositories, hill-climb rounds, anything the company owns	Company admin (Et)
Person 5	Claude Ultra ChatGPT Pro	Their own projects. Extra hill-climb capacity only when they start the round themselves	Person 5
Person 6	Claude Team premium seat ChatGPT Business premium seat	Company repositories, hill-climb rounds, anything the company owns	Company admin (Et)
Person 6	Claude Ultra ChatGPT Pro	Their own projects. Extra hill-climb capacity only when they start the round themselves	Person 6

Hardware and software

Why Linux

T3 Code runs on Linux as a systemd user service. With lingering turned on (`sudo loginctl enable-linger`), each person's service starts at boot and keeps running after they log out. On macOS the service stops at logout, so six people would need six logged-in desktop sessions. Linux also needs no screen, keyboard or desktop. Claude Code, Codex and git all run on Linux.

Phase 1, now: an Ubuntu VM on Et's MacBook Pro

- Run Ubuntu Server 24.04 in a VM (UTM is free). Every setup step then carries over to the tower unchanged.
- Turn off sleep on AC power: `sudo pmset -c sleep 0`. This needs Et's password.
- Free at least 80 GB for the VM, worktrees and test runs. The Mac had 7.8 GB free at the last audit.
- Add a backup drive. There was no backup destination at the last audit.
- Phase 1 runs one or two people at a time, and it is where the six-user test runs before any hardware is bought.

Phase 2: the Linux tower

Item	Why it is needed	Price	Status
AMD Ryzen 9 9950X, 16 cores	Sixteen cores for six worktrees building and testing at once	$549	check
64 GB DDR5 (two 32 GB sticks)	Six users, their agents and test runs in memory	$880	check
AM5 motherboard with 2.5 Gb Ethernet	Wired Ethernet, no Wi-Fi drops	$230	estimate
2 TB NVMe SSD	Room for six users, worktrees and caches	$200	estimate
Case, 850 W power supply and air cooler	Quiet enough to sit in an office	$250	estimate
UPS, 1500 VA line-interactive	Keeps rounds alive through short outages	$250	estimate
Backup drive, 8 TB	Local hourly backups with restic	$180	estimate
Gigabit switch and Ethernet cables	Stable link for remote sessions	$60	estimate
Total hardware		**$2,599**	

Memory prices were inflated in late 2026, so the CPU and memory lines are tagged "check". The Framework Desktop with 128 GB ($4,139 with extras) only makes sense if the team wants to run local models. The 32 GB Framework ($1,749) fits two or three people at a time. The Financials report compares all three.

- Encrypt the disk with LUKS and enroll the TPM chip (`systemd-cryptenroll --tpm2-device=auto`), so the tower restarts after a power cut without someone typing a password. The trade-off is that a stolen tower unlocks itself, so keep it in a locked room.
- Set the BIOS to power on after power loss.
- Turn on unattended security upgrades, and reboot on a schedule when no round is running.

Software

Software	Role	Cost
Ubuntu Server 24.04 LTS	Hub operating system, one Linux user per person	Free
T3 Code	Agent workspace for each person, run as that user's systemd service on its own port. Holds a Claude entry and a Codex entry per account, each with its own config folder and login.	Free, MIT license. T3 Connect is free for now, three machines per account
Claude Code CLI	Runs the Claude agents inside T3	Included in both account types
Codex CLI	Runs the GPT agents and the GPT reviewer	Included in ChatGPT Business and Pro
Git and GitHub	Worktrees per variant, pull requests for winners	Existing plan
Eval runner	A script that runs tests and the benchmark, then prints one score	Built in-house, one per project
restic	Hourly backups to the local drive, nightly copy offsite	Free. Offsite storage in the budget
Results ledger	A tab-separated file in the repository, one row per variant	Free
Tailscale	Fallback remote access	Standard plan, $8 per user per month. The free plan is non-commercial

Rollout

Step	Change	Pass test	Rollback
1	MacBook power, disk and backup fixes. Ubuntu Server VM installed	Mac stays awake 24 hours on AC. One backup completes. The VM survives a Mac restart	`sudo pmset -c sleep 1`, delete the VM
2	Create the Claude Team org and the ChatGPT Business workspace, two monthly premium seats each	Both people sign in, and Claude Code and Codex each report the company account	Remove seats before the first bill
3	One project's eval runner and ledger	Enough baseline runs to measure the noise margin. Every expected test runs	Delete the branch
4	First hill-climb round with two seats	Ledger has rows. The winner opens a pull request. GPT review is attached	Close the pull request
5	In the Ubuntu VM, create six Linux users with lingering, each running T3 on its own port	All six reach only their own service from a phone on cellular. Services survive logout, reboot and network loss	Delete the test users
6	Price the tower parts that week and buy them. Copy the VM's setup to the tower	Step 5 passed. One task's real CPU and memory use fits six at once	Return window

7	Twelve team seats on annual billing, full rounds	A week of rounds with no account limit hit before the budget stop	Drop K back to two

Prepared October 31, 2026 for Et. Prices tagged "verified" were read from the vendor's page on that date. Prices tagged "check" come from recent pricing guides because the vendor page blocked automated reads. Prices tagged "estimate" are placeholders that need a quote before purchase.